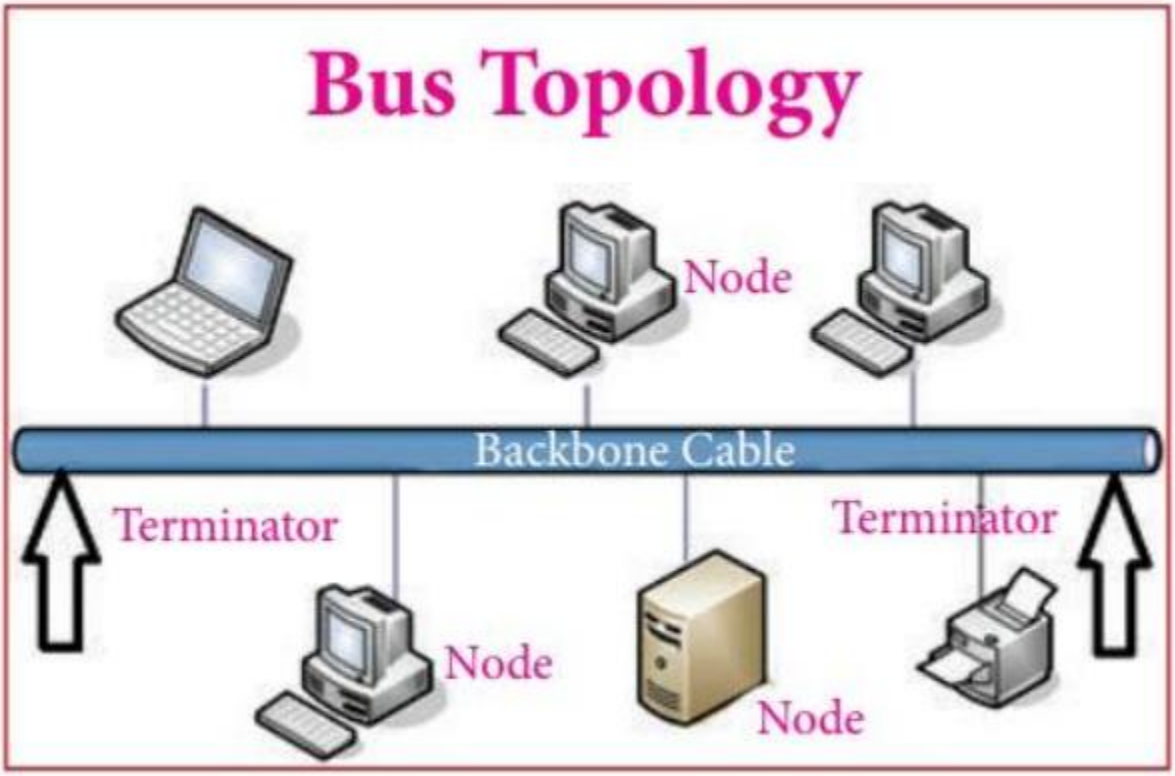
NETWORK TOPOLOGIES THEORY

1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be use



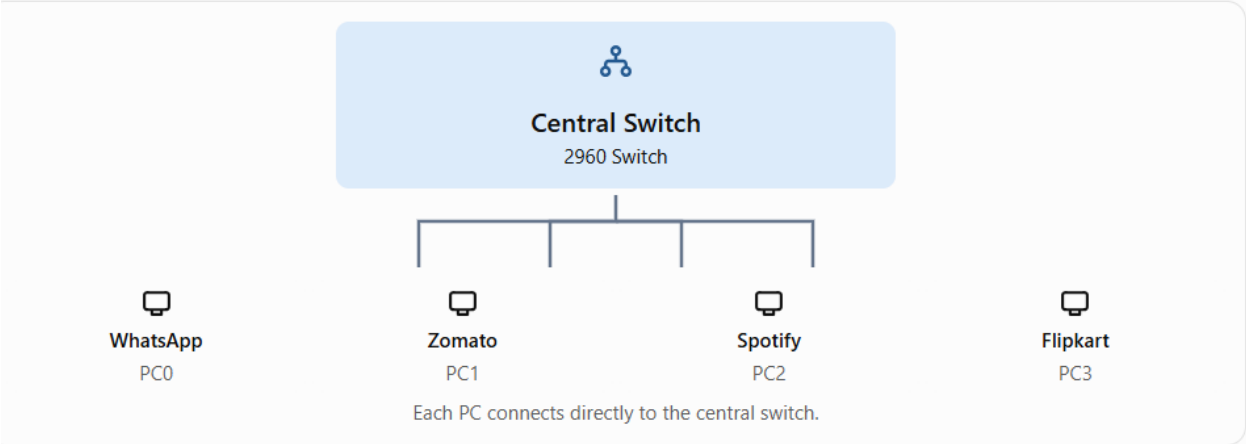
Real-world example: Older Ethernet networks using coaxial cable (10BASE2) used bus topology to connect multiple computers through one shared cable.

2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

Cisco Packet Tracer: Star Topology:-

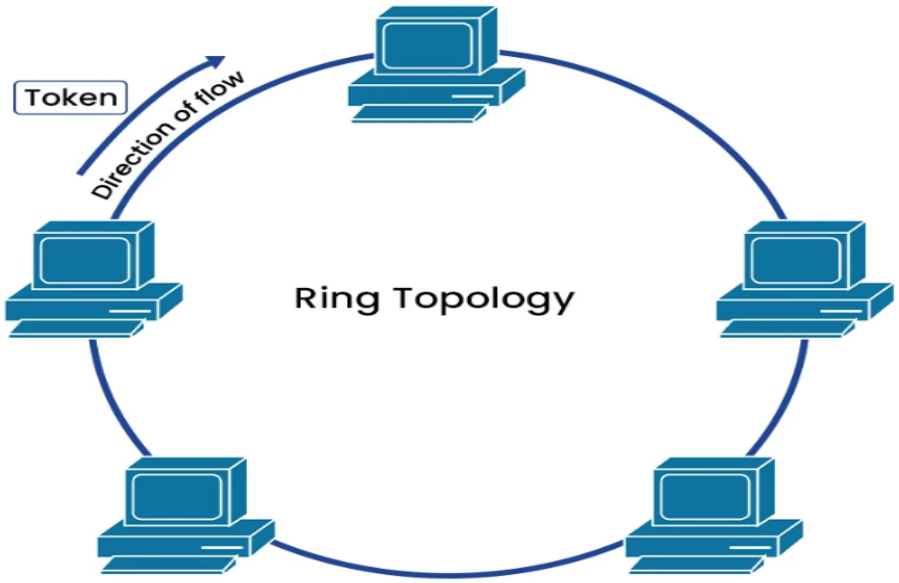
Use these device names:

- PC0 → WhatsApp
- PC1 → Zomato
- PC2 → Spotify
- PC3 → Flipkart
- Switch0 → Central Switch



3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Hint: Use arrows to show the direction of data flow.



Explanation:

1. In a ring topology, each device connects to two neighboring devices, forming a closed loop.
2. Data travels from one device to the next in the direction shown by the arrows until it reaches its destination.
3. In a traditional unidirectional ring, a break in the connection can disrupt communication unless a backup mechanism is available.

4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Comparison of Mesh and Star Topologies:-

Topology	Advantage	Disadvantage
Mesh Topology	1. Highly reliable because multiple paths may exist between devices. 2. Failure of one link usually does not stop the entire network.	1. Expensive due to many cables and ports. 2. Complex to install and maintain.
Star Topology	1. Easy to install and manage. 2. Failure of one device or cable usually does not affect other devices.	1. If the central switch fails, the network stops working. 2. Requires more cable than some simpler topologies, such as bus topology.